



BLOOD

Physiology, BSC MEDICAL TECHNOLOGY

BODY FLUIDS

Total water in the body is about 40L. it is distributed into two major compartments:

1. Intracellular fluid (ICF) : 22L forms 55% of the total body water
2. Extracellular fluid (ECF): 18L forms about 45% of the total body water.

ECF is divided into five subunits-

- i. Interstitial fluid and lymph (20%) ----->12L
- ii. Plasma (7.5%)----->2.75 L
- iii. Fluid in bones (7.5%)
- iv. Fluid in the dense connective tissue like cartilage (7.5%)
- v. Transcellular fluids (2.5%) that includes
 - a. Cerebrospinal fluid
 - b. Intraocular fluid
 - c. Digestive juices
 - d. Serous fluid
 - e. Synovial fluid in joints
 - f. Fluid in the urinary tract.

} ->3.25 L

COMPOSITION OF BODY FLUIDS

Organic substances - glucose, amino acids, fatty acids and other lipids, hormones and enzymes.

Inorganic substances - sodium, potassium, calcium, magnesium, chloride, bicarbonate, phosphate and sulphate.

MEASUREMENT OF BODY FLUID VOLUME

Indicator dilution method

A known quantity of a substance such as dye is administered into a specific body fluid compartments and allowed it to get mixed into the body. Then a sample of body fluid is extracted and concentration of the dye is calculated.

Interstitial fluid volume =

ECF volume - plasma volume

ICF volume =

Total fluid volume - ECF volume

TABLE 6.2: Marker substances used to measure body fluid compartments

Fluid compartment	Marker substances
Total body water	1. Deuterium oxide (D ₂ O) 2. Tritium oxide (T ₂ O) 3. Antipyrine
Extracellular fluid	1. Radioactive sodium, chloride, bromide, sulfate and thiosulfate. 2. Non-metabolizable saccharides like inulin, mannitol, raffinose and sucrose
Plasma	1. Radioactive iodine (¹³¹ I) 2. Evans blue (T-1824)

TONICITY OF THE FLUID

Movement of water between fluid compartments is not influenced by small molecules like urea and alcohol, rather the larger molecules such as sodium and glucose which crosses the cell membrane and influence the movement of water. Thus, the large molecules are called osmoles. Osmolarity which causes the

movement of water in between the compartment is called effective osmolarity and the effective osmoles responsible for this.

Tonicity is the measurement of effective osmolarity. In terms of tonicity, the solutions are divided into three categories-

1. *Isotonic solution* - the fluid and the body fluid has the same number of large molecules or effective osmoles. No movement of water molecules takes place in this type of solutions.
2. *Hypertonic solution* - fluid which has greater effective osmolarity than the body fluids is called hypertonic solution.
3. *Hypotonic solution* - fluid which has less effective osmolarity than the body fluids is called hypotonic solution.

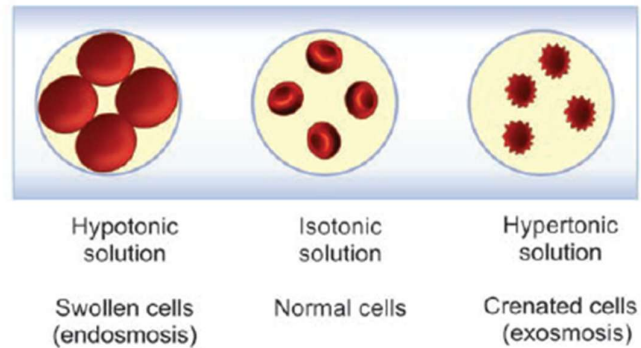


FIGURE 6.2: Effect of isotonic, hypertonic and hypotonic solutions on red blood cells

BLOOD

Blood is connective tissue in fluid form. it carries oxygen from lung to the tissues and CO_2 from tissue to the lungs

Properties -

- Blood is red in colour. Arterial blood is scarlet red because it contains oxygen and venous blood is purple red because of more carbon dioxide.
- Volume = 5L in males and 4.5L in females.
- Blood is slightly alkaline; pH of the blood is 7.4.
- Blood is five times viscous than the water and it is mainly because of the presence of RBC and plasma

COMPOSITION OF BLOOD

Blood contains -

- Blood cells which are called formed elements - 45%
- The liquid portion which is called plasma - 55%

FORMED ELEMENTS

There are three types of cells present in the blood

1. Red blood cells or erythrocytes
2. White blood cells or leukocytes
3. Platelets or thrombocyte

Hematocrit value

When blood is collected from hematocrit tube along with suitable anticoagulant and centrifuged for 30 minutes at a speed of 3000 revolutions per minute, the red blood cells settle down at the bottom having a clear plasma at the top. The volume

of red blood cells are measured in percentage is called hematocrit value. In between the layers of RBC and plasma there is a white buffy coat which is formed by aggregation of white blood cells and platelets.

PLASMA

Plasma is a straw-coloured clear liquid part of blood. It contains **91% to 92% of water** and 9% to 8% of solids. The solids are the organic and inorganic substances.

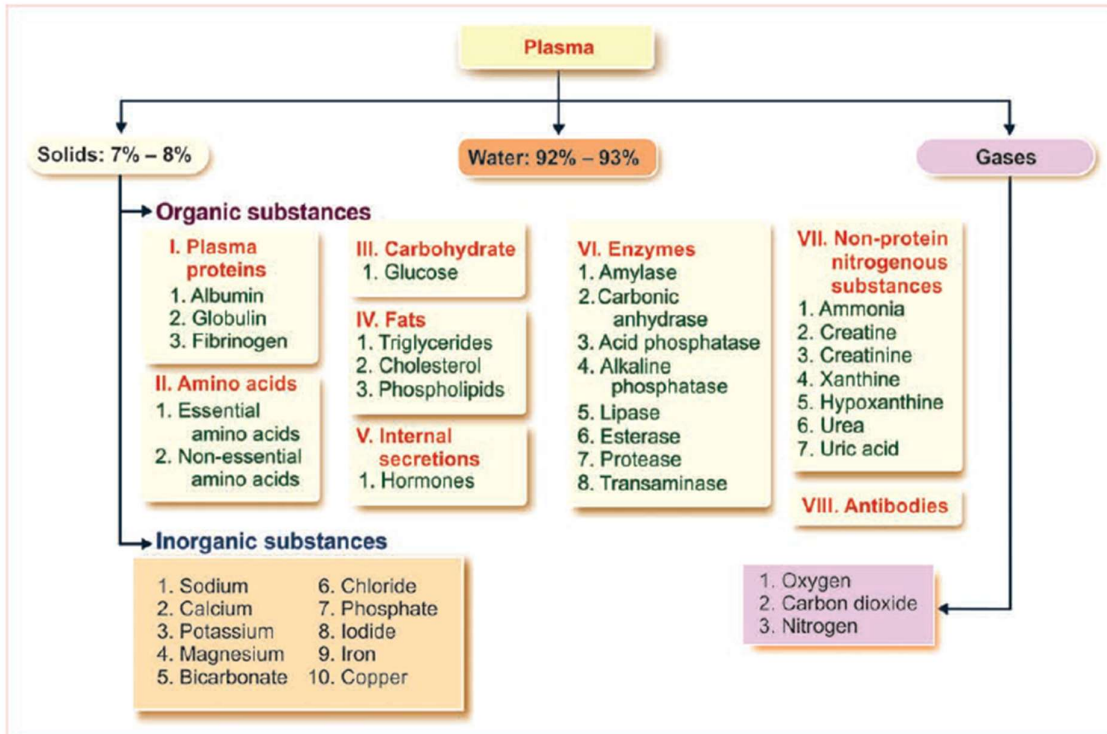


FIGURE 7.1: Composition of plasma

SERUM

Plasma without the clotting factors is called serum. It has same volume as plasma except the clotting factors.

Serum = plasma - fibrinogen

FUNCTION OF BLOOD

- **Nutritive function** - nutritive substances like glucose, amino acids, lipids and vitamins derived from digested food are absorbed from gastrointestinal tract and carried by blood to different parts of the body for growth and production of energy.
- **Respiratory function** - transport of the gases is done by blood.
- **Excretory function** - waste products formed in the tissues during various metabolic activities are removed by blood.
- **Transport of hormones and enzymes** - hormones which are secreted by ductless (endocrine) glands are released directly into the blood. The blood transports these hormones to their target organs/tissues. Blood also transports enzymes.

- **Regulation of water balance** - water content of the blood is freely interchangeable with interstitial fluid.
- **Regulation of acid-base balance** - plasma proteins and hemoglobin act as buffers and help in the regulation of acid-base balance.
- **Regulation of body temperature** - because of the high specific heat of the blood, it is responsible for maintaining thermoregulatory mechanism of the body.
- **Storage function** - water and some other important substances like proteins, glucose, sodium and potassium are stored in the blood and these are supplied by the blood during the conditions like starvation, fluid loss, electrolyte loss etc.
- **Defence mechanism** - while blood cells of the blood plays a very important role in the defence mechanism of the body.

PLASMA PROTEINS

Plasma proteins are-

1. Albumin
2. Globulin
3. Fibrinogen

Serum only contains albumin and globulin. Fibrinogen is absent in serum because it gets converted into fibrin during blood clotting.

Normal values - total protein : 7.3 g/dl

Serum albumin : 4.7 g/dl

Serum globulin : 2.3 g/dl

Fibrinogen : 0.3 g/dl

Normal ratio of albumin and globulin in the plasma is **2:1**

Methods of separating plasma proteins involves -

1. Precipitation method
2. Salting-out method
3. Electrophoretic method
4. Ultracentrifugation method
5. Gel filtration chromatography
6. Immunoelectrophoretic method

Properties

- **Oncotic pressure** -
 - Plasma proteins are responsible for the osmotic pressure in the blood. Osmotic pressure exerted by protein in the plasma is called oncotic pressure. Normally it is about **25mm of Hg**. **Albumin** plays an important role in exerting oncotic pressure.
- **Origin** -
 - In embryo - the plasma proteins are synthesized by the **mesenchyme cells**
 - In adults - the plasma proteins are synthesized by the **reticuloendothelial cells of liver**.

FUNCTIONS OF PLASMA PROTEINS

1. Role in coagulation of blood - **fibrinogen** helps in coagulation

2. Role in defence mechanism of the body -

Gamma globulins play an important role in the defence mechanism of the body by acting as antibodies. These proteins are also called immunoglobulins

3. Role in transport mechanism -

Plasma proteins are essential for the transport of various substances in the blood. **Albumin, alpha globulin, and beta globulin** are responsible for the transport of the hormones, enzymes etc.

PLASMAPHERESIS

Plasmapheresis is an experiment procedure done in animals to demonstrate the importance of plasma proteins.

In this experiment, blood from the dog (animal for experiment) is removed completely from the body. Red blood cells are separated from plasma and washed in saline and reinfused into the body of the same dog along with physiological solution called Locke's solution.

Due to sudden lack of proteins, the animal undergoes a state of shock and if the animal is fed again with diet containing sufficient high quality of proteins, the **liver** again synthesizes the plasma proteins and the dog restores its health again within seven days.

This experiment is done to demonstrate the following -

1. Importance of plasma proteins for survival
2. Synthesis of plasma proteins from the liver.

In clinical aspects, the plasmapheresis is used to treat autoimmune diseases. Since in **autoimmune disease** the antibodies fails to recognise and attacks the own tissue. Most of these antibodies are the plasma proteins which are washed away and resynthesized from liver by this method.

RED BLOOD CELLS

Red blood cells are also called erythrocytes. They are non-nucleated formed elements of the blood. They are red in colour due to the presence of colouring pigment called haemoglobin. RBCs play a vital role in the transport of respiratory gases.

Normal value - 5million per cubic mm of blood in adult males

4.5million per cubic mm of blood in adult females

NORMAL SHAPE

Normally RBCs are disk shaped and biconcave. Central portion is thinner and periphery is thicker. Advantages of biconcave shape of RBCs-

1. *Biconcave shape* helps in equal and rapid diffusion of oxygen and other substances into the interior of the cell
2. *Larger surface area* is provided for absorption or removal of different substances.
3. Minimal tension is offered on the membrane when the volume of the cell alters
4. Because of the biconcave shape, while passing through minute capillaries, RBCs squeeze through the capillaries very easily without getting damaged.

NORMAL SIZE

Diameter	:7.2 μ
Thickness	:at the periphery it is thicker with 2.2 μ and at the center it is 1 μ
Surface area	:120 sq. μ
Volume	:85 to 90 cu μ

LIFE SPAN OF RBC

Average life span of RBC is about **120 days**. After the lifetime the senile (old) RBCs are destroyed in the **reticuloendothelial system**.

Lifespan of RBC is determined by radioisotopes method. RBCs are tagged with radioactive

substances like **radioactive chromium and radioactive iron**. Lifespan of RBC is determined by studying the rate of **loss of radioactive cells** from circulation

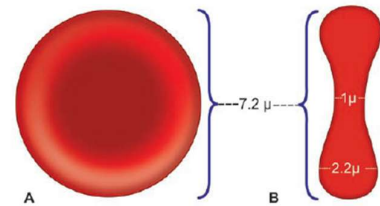


FIGURE 9.1: Dimensions of RBC. A. Surface view, B. Sectioned view.

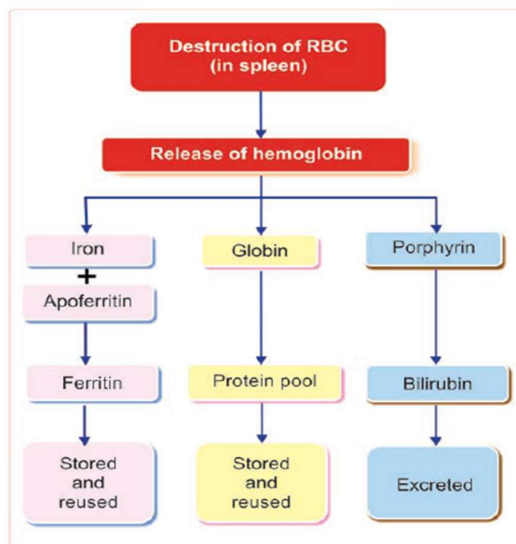


FIGURE 9.3: Fate of RBC

When the cells becomes older (120 days) the cell membrane becomes more fragile. The diameter of capillaries is less or equal to the RBC thus younger cells can squeeze through it. *But the senile (older) RBCs are destroyed while squeezing through because of its fragile nature.* The destruction occurs mainly in the *capillaries of red pulp of spleen because diameter of splenic capillaries are very less. So, spleen is called the graveyard of RBCs.*

Destroyed RBCs are fragmented and hemoglobin is released from fragmented parts. Hemoglobin is phagocytized by the macrophages present in the liver (Kupffer cells), spleen and bone marrow.

FUNCTIONS OF RBCs**1. Transport of oxygen from the lungs to the tissues**

Hemoglobin in RBC combines with oxygen to form oxyhemoglobin. About 97% of oxygen is transported in blood in the form of oxyhemoglobin.

2. Transport of carbon dioxide from the tissue to the lungs

Hemoglobin combines with carbon dioxide to form carbhemoglobin. About 30% of carbon dioxide is transported in this form. about 63% of the carbon dioxide is transported in bicarbonate form in the RBC

3. Buffering action in blood

Hemoglobin functions as a good buffer. By this action it regulates the hydrogen ion concentration and thereby plays a major role in maintaining acid-base balance

4. In blood group determination

RBCs carry the blood group antigens like A antigen, B antigen and RH factor. This helps in the determination of blood groups and enables to prevent reactions due to incompatible blood transfusion.

VARIATIONS IN NUMBER OF RED BLOOD CELLS

PHYSIOLOGICAL VARIATIONS

A. Increased RBC count

Increased RBC count is known as polycythemia. It occurs in the following conditions -

1. **Age** - at birth, the RBC count is 8 to 10 million/cu mm of blood. The count decreases within 10 days after the birth due to destruction of RBCs causing physiological jaundice in some newborn babies.

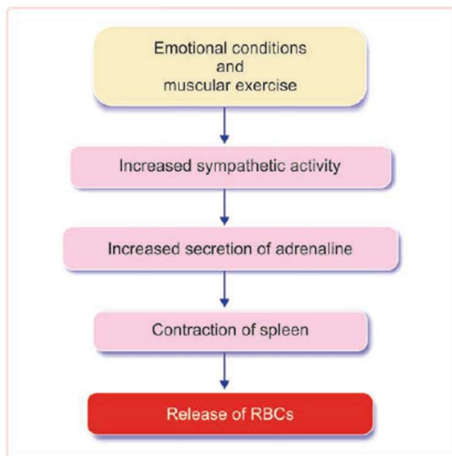


FIGURE 9.5: Physiological polycythemia in emotional conditions and exercise

2. **Sex** - before puberty and after menopause in females the RBC count is similar to that in males. During reproductive period of females, the count is less than that of males.

3. **High altitude** - inhabitants of mountains have increased RBC count and it is due to hypoxia (decreased oxygen supply to the tissue). Hypoxia stimulates kidney to secrete erythropoietin which in turn stimulates bone marrow to produce more RBC.

4. **Muscular exercise** - there is a temporary increase in the count of RBC during exercise due to mild hypoxia and contraction of spleen. Hypoxia increases sympathetic activity resulting in secretion of adrenaline from adrenal medulla. Adrenaline contracts the spleen and the stored RBC are released into the blood.

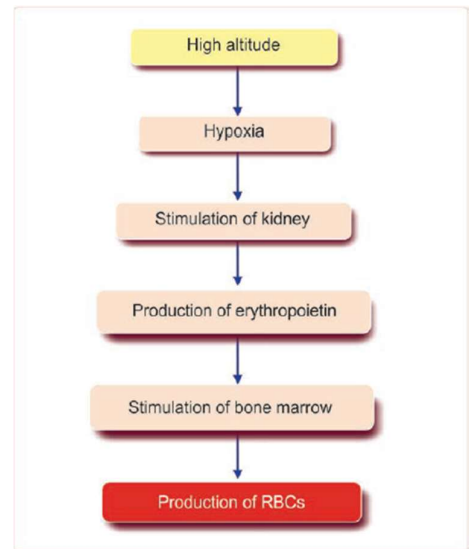


FIGURE 9.4: Physiological polycythemia in high altitude

B. Decreased RBC count

Decreased RBC count occurs in the following conditions -

1. **High barometric pressure** - at high barometric pressure as in the deep sea, when oxygen tension in the blood is higher, the RBC count decreases.
2. **During sleep** - all the activities of the body are decreased during the sleep including the production of RBC and hence during sleep, the RBC count decreases.
3. **Pregnancy** - in pregnancy, the RBC count decreases, it is because of the increase in ECF volume. Increased in ECF volume increases the plasma volume resulting in hemodilution.

ERYTHROPOESIS - The process of synthesizing erythrocytes is called erythropoiesis.

SITE OF ERYTHROPOIESIS

In fetal life

1. *Mesoblastic stage* - during the 1st two months of intrauterine life, the RBC are produced from mesenchyme of yolk sac
2. *Hepatic stage* - from the third month of intrauterine life, the liver is the main organ that produces RBC. Spleen and lymphoid organs are also involved in erythropoiesis.
3. *Myeloid stage* - during the last three months of intrauterine stage, the RBCs are produced from red bone marrow and liver

In newborn babies, children and adults

1. Up to the age of 20 years - RBCs are produced from red bone marrow of all bones (long bones and all the flat bones)
2. After the age of 20 years - the RBCs are produced from membranous bones like vertebra, sternum, ribs, scapula, iliac bones and skull bones and ends of the long bones. After the age of 20 years the **shaft of the long bones becomes yellow due to deposition of fats and thus it loses the erythropoietic function.** Liver and spleen may also produce RBC if red bone marrow is destroyed or fibrosed.

PROCESS OF ERYTHROPOIESIS

Hemopoietic Stem cell are the primary cells for the formation of all the blood cells. Hemopoietic stem cells are called uncommitted pluripotent hemopoietic stem cells (PHSC). When these cells are designated to form a particular type of blood cells, then this cells are known as committed PHSC. Committed PHSC are of two types -

1. Lymphoid stem cells
2. Colony forming blastocyte

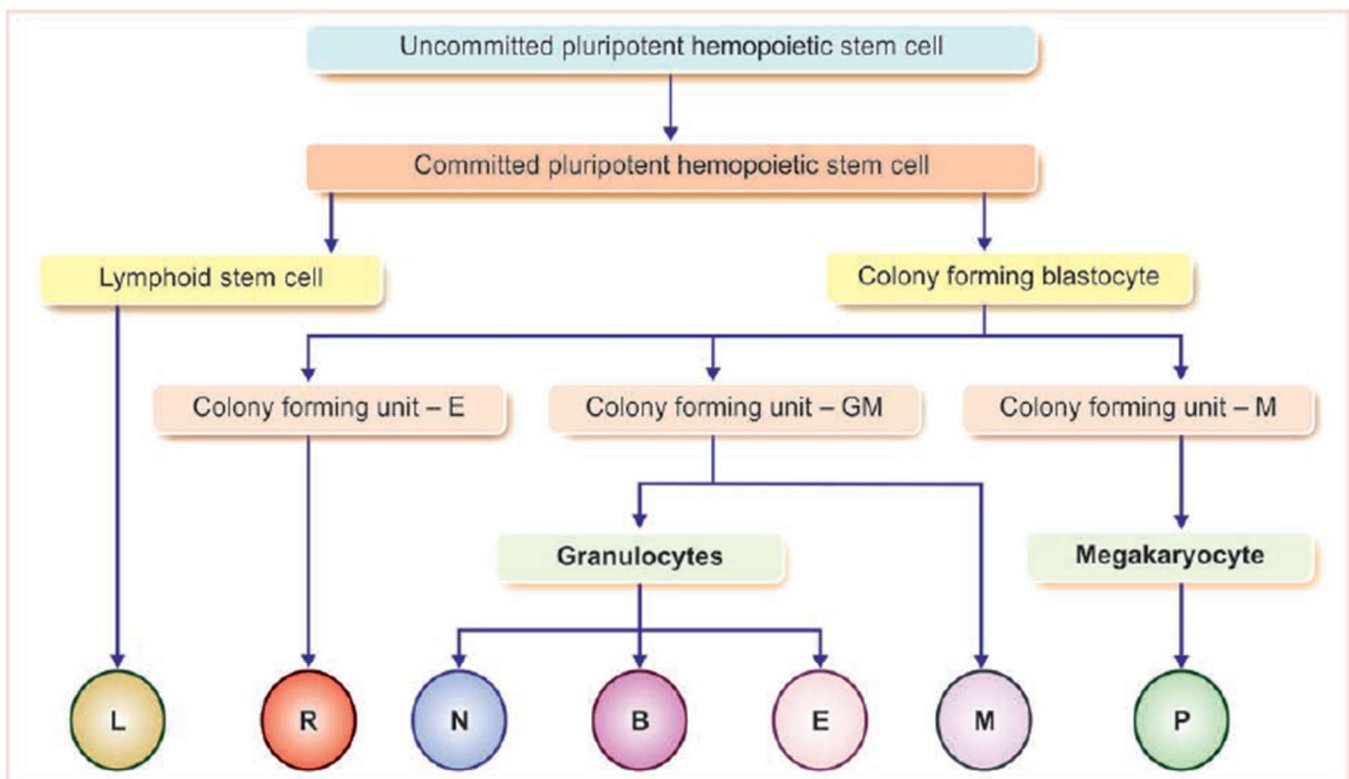


FIGURE 10.1: Stem cells. L = Lymphocyte, R = Red blood cell, N = Neutrophil, B = Basophil, E = Eosinophil, M = Monocyte, P = Platelet.

- a. CFU-E (colony forming unit-erythrocyte)
- b. CFU-GM(colony forming unit-granulocyte/monocyte)
- c. CFU-M (colony forming unit-megakaryocytes)

STAGES OF ERYTHROPOIESIS

1. Proerythroblast - proerythroblast or megaloblast is the first cell derived from CFU-E. it has very large diameter and nucleus is large almost occupies the most of the cell space completely. **Synthesis of hemoglobin starts** in this stage.
2. Early normoblast - early normoblast has small diameter than proerythroblast and nucleoli disappears. Condensation of chromatin network occurs. The cytoplasm is basophilic in nature
3. Intermediate normoblast - cell is smaller than the early normoblast and nucleus is still present but chromatin network shows further condensation. Appearance of hemoglobin takes place at this stage. Cytoplasm is already basophilic but due to **appearance of hemoglobin**, it stains with both acidic as well as basic dye.
4. Late normoblast - diameter of the cell decreases further and nucleus becomes very small with very much condensed chromatin network and it is known as ink spot nucleus. Quantity of hemoglobin increases and the cytoplasm is almost acidophilic. In the final stage of late normoblast just before the next stage of erythropoiesis, the nucleus disintegrates and disappears. The process by which the nucleus disappears is called pyknosis.
5. Reticulocytes - the cells are slightly larger than the mature RBC and contains reticular network of reticulum which is formed by the remnants of disintegrated organelles. Reticulocytes are basophilic in nature due to the presence of remnants of Golgi apparatus, mitochondria and other organelles of cytoplasm. During this stage the cells enters into the blood capillaries through capillary membrane from the site of production by diapedesis.
6. Mature erythrocytes - reticular network disappears and the cell becomes mature RBC and attains biconcave shape. The cell decreases in size to 7.2 μm . the mature RBC with hemoglobin and without nucleus.

It requires 7days for the development and maturation of RBC from proerythroblast. It requires 5days up to reticulocytes and 2days for the maturation of RBC

TABLE 10.1: Important events during erythropoiesis

Stage of erythropoiesis	Important event
Proerythroblast	Synthesis of hemoglobin starts
Early normoblast	Nucleoli disappear
Intermediate normoblast	Hemoglobin starts appearing
Late normoblast	Nucleus disappears
Reticulocyte	Reticulum is formed. Cell enters capillary from site of production
Matured RBC	Reticulum disappears Cell attains biconcavity

FACTORS AFFECTING ERYTHROPOIESIS

GENERAL FACTORS

1. *Erythropoietin* - most important general factor for the erythropoiesis. Erythropoietin is secreted from peritubular capillaries of the kidney and a small quantity from brain and liver. Hypoxia is the stimulant for the secretion

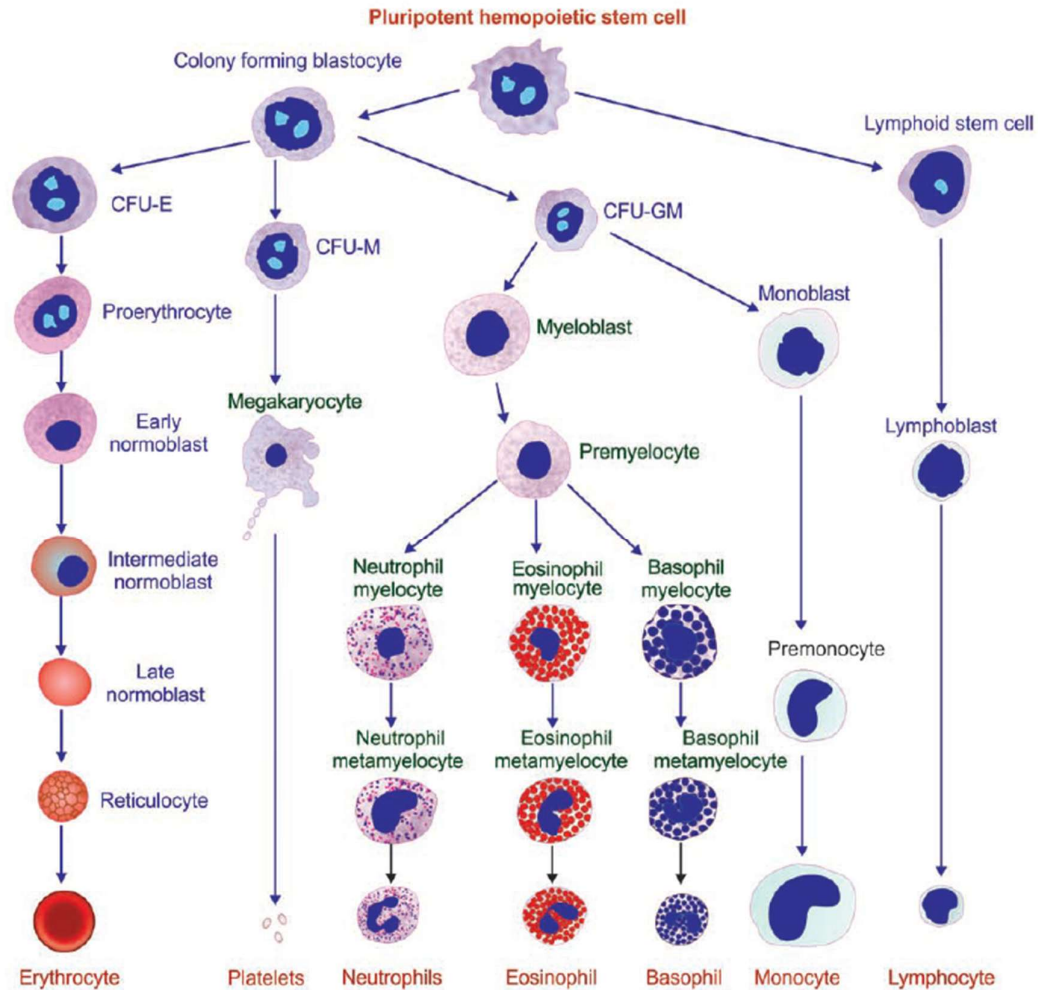


FIGURE 10.2: Stages of erythropoiesis. CFU-E = Colony forming unit-erythrocyte, CFU-M = Colony forming unit-megakaryocyte, CFU-GM = Colony forming unit-granulocyte/monocyte.

of erythropoietin. Erythropoietin causes formation and release of new RBCs into the circulation.

2. *Thyroxin* - being a general metabolic hormone, thyroxin accelerates the process of erythropoiesis at many different levels. So, hyperthyroidism and polycythemia are common.
3. *Vitamins* - some vitamins are necessary for the process of erythropoiesis. Deficiency of these vitamins causes anemia associated with other disorders.

MATURATION FACTORS

1. *Vitamin B12 (cyanocobalamin)* - vitamin B12 is essential for synthesis of DNA in RBCs. Its deficiency leads to failure and maturation of the cell and reduction in the cell division. Deficiency of vitamin B12 causes pernicious anemia.
2. *Intrinsic factor of castle* - intrinsic factor of the castle is produced by the gastric mucosa of the parietal cells of the gastric glands. It is essential for the absorption of vitamin B12 from intestine. In the absence of intrinsic factor, vitamin B12 is not absorbed from intestine and it leads to the pernicious anemia. Deficiency of the intrinsic factor occurs in:
 - a. Severe gastritis
 - b. Ulcer
 - c. Gastrectomy
3. *Folic acid* - folic acid is also essential for maturation of RBC since it is required for DNA synthesis in the blood cell. Failure of DNA synthesis results in failure of maturation and it leads to anemia. Anemia caused by folic acid deficiency is called megaloblastic anemia

FACTORS NECESSARY FOR HEMOGLOBIN FORMATION

1. *First class proteins and amino acids* - proteins of high biological value are essential for the formation of hemoglobin. Amino acids derived from these proteins are required for the synthesis of protein part of hemoglobin, i.e. the globin.
2. *Iron* - necessary for the formation of the heme part of the hemoglobin.
3. *Copper* - necessary for the absorption of iron from the gastrointestinal tract.
4. *Cobalt and nickel* - these metals are essential for the utilisation of iron during hemoglobin formation
5. *Vitamins* - vitamin C, riboflavin, nicotinic acid and pyridoxine are also essential for the formation of hemoglobin

HEMOGLOBIN

Hemoglobin is the iron containing coloring matter of red blood cells. It is a chromoprotein forming 95% of the dry wet of the RBC. Normal content of hemoglobin are as follows -

- | | |
|---------------------|-----------|
| 1. In adult males | 15 g/dl |
| 2. In adult females | 14.5 g/dl |

Function Of Hemoglobin

1. Transport of oxygen
2. Transport of carbon dioxide
3. Buffer action

Structure of hemoglobin

Hemoglobin is a conjugated protein. It consists of a protein combined with an iron-containing pigment. The protein part is globin and the iron-containing pigment is heme -

- *Iron* - normally it is present in the ferrous form (Fe^{2+}) form. it is an unstable and loose form. in some unstable conditions, the iron is converted into ferric state which is stable state.
- *Porphyrin* - the pigment part of heme is called porphyrin. It is formed by four pyrrole rings called, I, II, III, IV. The pyrrole rings are attached to one another by methane (CH_4) bridges. There is iron attached to 'N' of each pyrrole rings and 'N' of each globin molecule.
- *Globin* - globin contains four peptide chains. Two of them are β chains and two are α chains

ABNORMAL HEMOGLOBIN DERIVATIVES

CARBOXYHEMOGLOBIN

Carboxyhemoglobin or carbon monoxyhemoglobin is the abnormal hemoglobin derivative formed by the combination of carbon monoxide with hemoglobin. Carbon monoxide is colorless and odourless gas. Since hemoglobin has 200 times more affinity for carbon monoxide than oxygen, it hinders the transport of oxygen resulting in tissue hypoxia. Sources of carbon monoxide are -

- Charcoal burning
- Coal mines
- Deep wells
- Underground drainage system
- Exhaust of gasoline engine
- Gases from guns and other weapons
- Smoke from fire
- Tobacco smoking

Signs and symptoms of carbon monoxide poisoning -

1. While breathing air with less than 1% of CO, the Hb saturation is 15% to 20% and mild symptoms like headache and nausea appear.
2. While breathing air with more than 1% of CO, the Hb saturation is 30% to 40%. It causes severe symptoms like-
 - a. Convulsions
 - b. Cardiorespiratory arrest
 - c. Unconsciousness and coma
3. When Hb saturation increases above 50%, death occurs.

ANEMIA

Anemia is the blood disorder, characterized by the reduction in -

1. Red blood cells count
2. Hemoglobin content
3. Packed cell volume

Generally, reduction in RBC count, hemoglobin content and PCV occurs because of

1. Decreased production of RBC
2. Increased destruction of RBC
3. Excess loss of blood from the body

CLASSIFICATION OF ANEMIA

Anemia is classified by two methods -

1. Morphological classification - depends upon the size and color of RBC
2. Etiological classification- on the basis of etiology (study of cause and origin)

MORPHOLOGICAL CLASSIFICATION

1. Normocytic normochromic anemia - size and color of RBC is normal but the number of RBC is less
2. Macrocytic normochromic anemia - RBCs are larger in size with normal color. RBC count is less
3. Macrocytic hypochromic anemia - RBCs are larger in size, mean corpuscular hemoglobin concentration is less which determines the color of RBC, so the cells are pale (less colored)
4. Microcytic hypochromic anemia - RBCs are smaller in size and less color

ETIOLOGICAL CLASSIFICATION

Etiologically anemia is divided into five types -

1. *Hemorrhagic anemia* - hemorrhage refers to the excessive loss of blood. Anemia due to hemorrhage is known as hemorrhagic anemia. It occurs in both acute and chronic hemorrhage conditions
 - a. *Acute hemorrhage* - acute hemorrhage refers to sudden loss of large quantity of blood as in the case of accident. Within about 24 hours after the hemorrhage, the plasma portion of the blood is replaced. However, the replacement of RBCs does not occur quickly and it takes at least 4 to 6 weeks. So, with less number of RBC, hemodilution occurs. However morphologically the RBCs are normocytic and normochromic. Decreased RBC count causes hypoxia, which stimulates the bone marrow to produce more number of RBC. So, the condition is corrected within 4 to 6 weeks.
 - b. *Chronic hemorrhage* - it refers to the loss of blood due to internal and external bleeding, over a long period of time. It occurs in conditions like peptic ulcer, purpura, hemophilia and menorrhagia. Due to continuous loss of blood, lot of iron is lost from the body causing iron deficiency. This affects the synthesis of hemoglobin resulting in less hemoglobin content in the cells. The cells also become small. Hence, the RBCs are microcytic and hypochromic.
2. *Hemolytic anemia* - hemolysis means destruction of RBCs. Anemia due to excessive hemolysis which is not compensated by increased RBC production is called hemolytic anemia. It is divided into two types :
 - a. *Extrinsic hemolytic anemia* - it is the type of anemia caused by destruction of RBCs from external factors. Healthy RBCs are hemolyzed by factors outside the blood cells such as antibodies, chemicals, and drugs. Extrinsic hemolytic anemia is also called autoimmune hemolytic anemia. Common causes of external hemolytic anemia :
 - i. Liver failure
 - ii. Renal disorder

- iii. Hypersplenism
 - iv. Burns
 - v. Infections like hepatitis, malaria and septicemia
 - vi. Drugs such as penicillin, antimalarial drugs and sulpha drugs
 - b. *Intrinsic hemolytic anemia* - It is the type of anemia caused by destruction of RBC because of the defective RBCs. This is because of production of unhealthy RBCs, which are short lived and are destroyed soon. Intrinsic hemolytic anemia is often inherited and includes sickle cell anemia and thalassemia
3. *Nutrition deficiency anemia* - anemia that occurs due to deficiency of a nutritive substance necessary for erythropoiesis is called nutrition deficiency anemia. The substances which are necessary for erythropoiesis are iron, protein and vitamins like C, B12 and folic acid. The types of nutrition deficiency anemia are :
- a. *Iron deficiency anemia* - most common type of anemia due to inadequate availability of iron for hemoglobin synthesis. RBCs are microcytic and hypochromic. Cause of iron deficiency anemia are :
 - i. Loss of blood
 - ii. Decreased intake of iron
 - iii. Poor absorption of iron from intestine
 - iv. Increased demand for iron in conditions like growth and pregnancy
 - b. *Protein deficiency anemia* - due to deficiency of proteins, the synthesis of hemoglobin is reduced. The RBCs are macrocytic and hypochromic
 - c. *Pernicious anemia or Addison's anemia* - due to the deficiency of vitamin B12. It is also called Addison's anemia. It is due to atrophy of the gastric mucosa because of autoimmune destruction of parietal cells. The gastric atrophy results in decreased production of intrinsic factor and poor absorption of vitamin B12, which is maturation factor for RBC. Synthesis of hemoglobin is almost normal in this type of anemia hence the RBC are macrocytic and normochromic
 - d. *Megaloblastic anemia* - megaloblastic anemia is due to deficiency of another maturation factor which is folic acid. Hence the RBCs are not matured. The RBCs are macrocytic and hypochromic
4. *Aplastic anemia* - aplastic anemia is due to the disorder of red bone marrow. Red bone marrow is reduced and replaced by fatty tissues. Bone marrow disorder occurs in the following conditions:
- i. Repeated exposure to X-ray or gamma radiation
 - ii. Presence of bacterial toxins, quinine, gold salts, benzene, radium etc.
 - iii. Tuberculosis
 - iv. Viral infections like hepatitis and HIV infections
- In aplastic anemia, the RBCs are normocytic and normochromic
5. *Anemia of chronic diseases* - anemia of chronic diseases is the second common type of anemia. It is characterized by the short life span of RBCs, caused by the disturbance in iron metabolism or resistance to erythropoietin

action. RBCs are normocytic and normochromic. Common causes anemia of chronic diseases:

- i. Non-infectious inflammatory diseases such as rheumatoid arthritis
- ii. Chronic infections like tuberculosis (infection caused by mycobacterium tuberculosis) and abscess (collection of pus in the infected tissue) in the lungs.
- iii. Chronic renal failure, in which the erythropoietin secretion decreases
- iv. Neoplastic disorder such as Hodgkin's disease and cancer of lung and breast.

SIGNS AND SYMPTOMS OF ANEMIA

1. *Skin and mucous membrane* - color of the skin and mucous membrane becomes pale. Paleness is more constant and prominent in buccal and pharyngeal mucous membrane. Skin loses the elasticity and becomes thin and dry. Thinning, loss and early greyness of hair occur. The nails becomes brittle and easily breakable.
2. *Cardiovascular system* - there is an increase in heart rate (tachycardia) and cardiac output. Heart is dilated and cardiac murmurs are produced.
3. *Respiration* - there is an increase in rate and force of respiration. Sometimes it leads to breathlessness and dyspnea (difficulty in breathing). Oxygen-hemoglobin dissociated curve is shifted to right.
4. *Digestion* - anorexia, nausea, vomiting, abdominal discomfort and constipation are common. In pernicious anemia, there is atrophy of papillae in tongue
5. *Metabolism* - basal metabolic rate increases in severe anemia.
6. *Kidney* - renal function is disturbed. Albuminuria is common
7. *Reproduction system* - in females, the menstrual cycle is disturbed. There may be menorrhagia, oligomenorrhea or amenorrhea.
8. *Neuromuscular system* - common neuromuscular systems are increased sensitivity to cold, headache, lack of concentration, restlessness, irritability, dizziness or vertigo and fainting. Muscle becomes weak and patient feels lack of energy and fatigued quite often quite easily.

WHITE BLOOD CELLS

White blood cells are colorless and nucleated formed elements of the blood. They are also known as leukocytes. White blood cells are larger in size as compared to RBCs and their role is in defence mechanism of the body and protect the body from invading organisms.

CLASSIFICATION OF WBC

Some of the WBCs have granules in the cytoplasm. Based on the presence or absence of granules in the cytoplasm, the leukocytes are divided into two categories

1. *Granulocytes* which have granules
2. *Agranulocytes* which don't have granules in the cytoplasm

Granulocytes

Depending upon the staining property of the granules, the granulocytes are classified into three categories -

- a. *Neutrophils* with granules taking both acidic and basic stains.
- b. *Eosinophils* with granules taking acidic stain
- c. *Basophils* with granules taking basic stain

Agranulocytes

Agranulocytes have plain cytoplasm without granules. Agranulocytes are of two types :

- a. *Monocytes*
- b. *Lymphocytes*

NEUTROPHILS - Neutrophils are also known as polymorphs have fine and small granules in the cytoplasm. The granules takes both acidic and basic stains. When stained with Leishman's stain (which contains acidic eosin and basic methylene blue) the granules appear violet in color. Nucleus is multilobed. And it has 2-5 lobes.

EOSINOPHILS - Eosinophils has coarse (large) granules in the cytoplasm, which stains pink or red with eosin. Nucleus is bilobed and spectacle shaped.

DIFFERENCE BETWEEN RBC AND WBC

TABLE 16.1: Differences between WBCs and RBCs

Feature	WBCs	RBCs
Color	Colorless	Red
Number	Less: 4,000 to 11,000/cu mm	More: 4.5 to 5.5 million/cu mm
Size	Larger Maximum diameter = 18 μ	Smaller Maximum diameter = 7.4 μ
Shape	Irregular	Disk-shaped and biconcave
Nucleus	Present	Absent
Granules	Present in some types	Absent
Types	Many types	Only one type
Lifespan	Shorter $\frac{1}{2}$ to 15 days	Longer 120 days

TABLE 16.3: Normal values of different WBCs

WBC	Percentage	Absolute value per cu mm
Neutrophils	50 to 70	3,000 to 6,000
Eosinophils	2 to 4	150 to 450
Basophils	0 to 1	0 to 100
Monocytes	2 to 6	200 to 600
Lymphocytes	20 to 30	1,500 to 2,700

Percentage of abundance in descending order:

Never **L**et **M**onkey **E**at **B**anana

BASOPHILS - have coarse granules in the cytoplasm. The granules stain purple blue with methylene blue. Nucleus is bilobed.

MONOCYTES - monocyte are the largest leukocytes. The cytoplasm is clear without granules. Nucleus is round, oval and horseshoe shaped, bean shaped or kidney shaped. Nucleus is either at the centre or pushed towards one side of the cell.

LYMPHOCYTES - like monocytes, lymphocytes also do not have granules in the cytoplasm. Nucleus is oval, bean shaped or kidney shaped. The nucleus occupies the whole of the cytoplasm. Depending upon the function, the lymphocytes are divided into two types -

1. T lymphocytes : cells concerned with cellular immunity.
2. B lymphocytes : cells concerned with humoral immunity

NORMAL WBC COUNT

Total WBC count - 4000 to 11000/cu mm of blood.

PATHOLOGICAL VARIATIONS IN THE WBC COUNT

Leukocytosis - leukocytosis is the increase in total leukocytes. It occurs in the conditions like -

- a. Infections
- b. Allergy
- c. Common cold
- d. Tuberculosis
- e. Glandular fever

Leukemia - leukemia is the condition which is characterised by abnormal

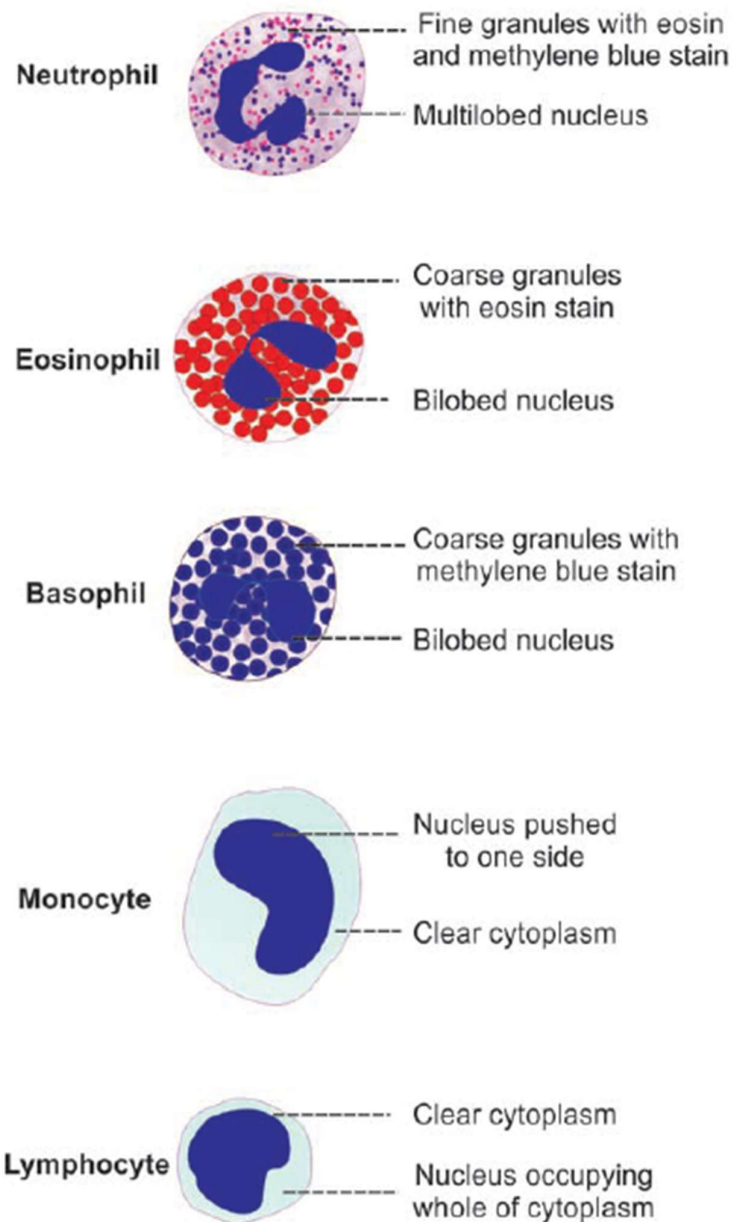


FIGURE 16.1: Different white blood cells

TABLE 16.2: Diameter and lifespan of WBCs

WBC	Diameter (μ)	Lifespan (days)
Neutrophils	10 to 12	2 to 5
Eosinophils	10 to 14	7 to 12
Basophils	8 to 10	12 to 15
Monocytes	14 to 18	2 to 5
Lymphocytes	7 to 12	½ to 1

and uncontrolled increase in leukocyte count more than 1000,000/cu mm. it is also called blood cancer.

Leukopenia - leukopenia is the decrease in the total WBC count. It occurs in the following physiological conditions -

- a. Anaphylactic shock
- b. Cirrhosis of liver
- c. Pernicious anemia
- d. Typhoid and paratyphoid
- e. Disorder of spleen
- f. Viral infections

TABLE 16.4: Pathological variations in different types of WBCs

Disorder	Variation	Conditions
Neutrophilia or neutrophilic leukocytosis	Increase in neutrophil count	1. Acute infections 2. Metabolic disorders 3. Injection of foreign proteins 4. Injection of vaccines 5. Poisoning by chemicals and drugs like lead, mercury, camphor, benzene derivatives, etc. 6. Poisoning by insect venom 7. After acute hemorrhage
Neutropenia	Decrease in neutrophil count	1. Bone marrow disorders 2. Tuberculosis 3. Typhoid 4. Autoimmune diseases
Eosinophilia	Increase in eosinophil count	1. Allergic conditions like asthma 2. Blood parasitism (malaria, filariasis) 3. Intestinal parasitism 4. Scarlet fever
Eosinopenia	Decrease in eosinophil count	1. Cushing's syndrome 2. Bacterial infections 3. Stress 4. Prolonged administration of drugs like steroids, ACTH and epinephrine
Basophilia	Increase in basophil count	1. Smallpox 2. Chickenpox 3. Polycythemia vera
Basopenia	Decrease in basophil count	1. Urticaria (skin disorder) 2. Stress 3. Prolonged exposure to chemotherapy or radiation therapy
Monocytosis	Increase in monocyte count	1. Tuberculosis 2. Syphilis 3. Malaria 4. Kala-azar
Monocytopenia	Decrease in monocyte count	Prolonged use of prednisone (immunosuppressant steroid)
Lymphocytosis	Increase in lymphocyte count	1. Diphtheria 2. Infectious hepatitis 3. Mumps 4. Malnutrition 5. Rickets 6. Syphilis 7. Thyrotoxicosis 8. Tuberculosis
Lymphocytopenia	Decrease in lymphocyte count	1. AIDS 2. Hodgkin's disease (cancer of lymphatic system) 3. Malnutrition 4. Radiation therapy 5. Steroid administration

PROPERTIES OF THE WHITE BLOOD CELLS

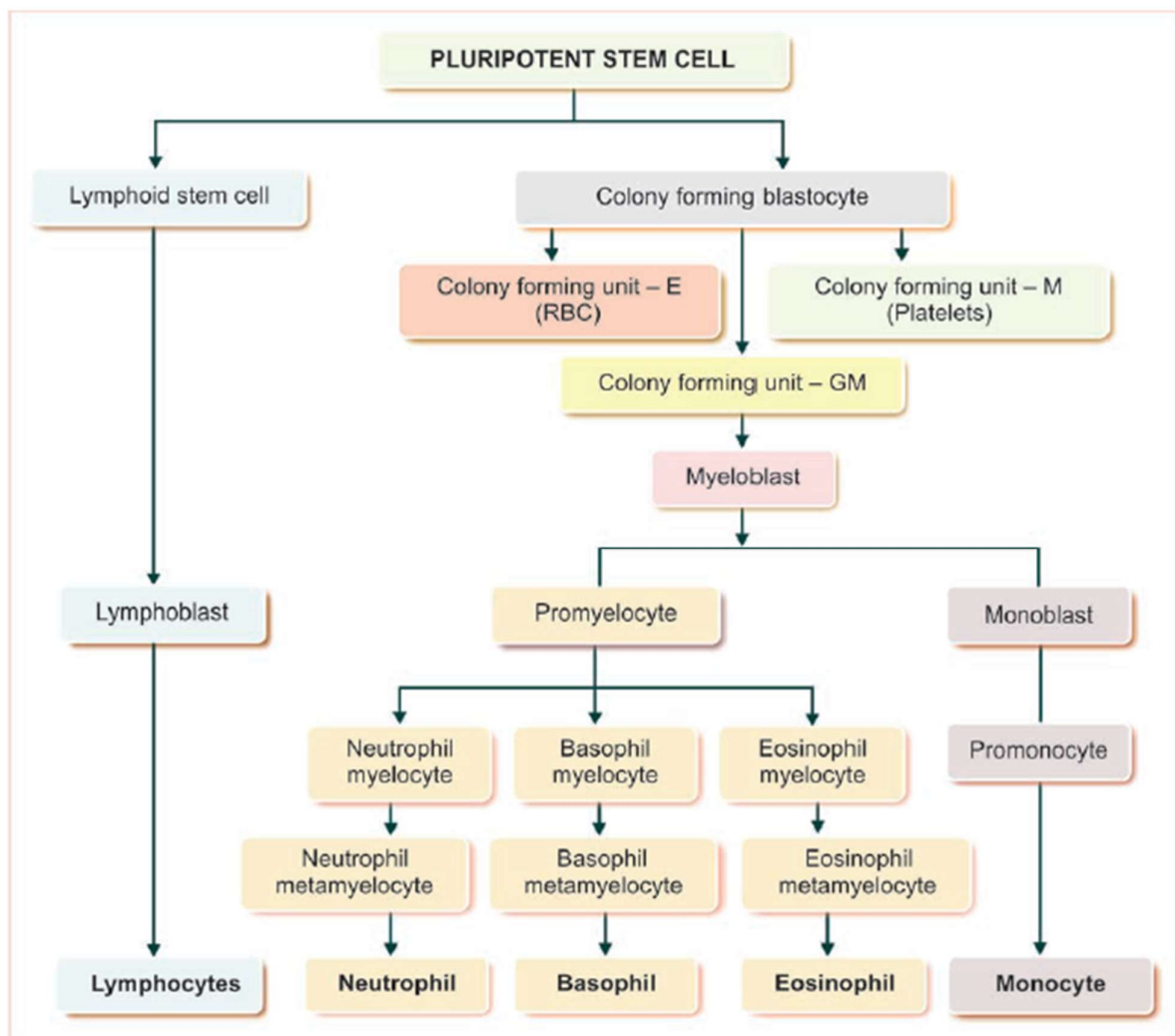
1. *Diapedesis* - diapedesis is the process by which the leukocytes squeeze through the narrow blood vessels
2. *Ameboid movement* - neutrophils, monocytes and lymphocytes shows ameboid movement characterised by protrusion of the cytoplasm and change in the shape.
3. *Chemotaxis* - chemotaxis is the attraction of WBCs towards the injured tissues by chemical substances released at the site of injury.
4. *Phagocytosis* - neutrophils and monocytes engulf the foreign bodies by the means of phagocytosis.

TABLE 16.3: Normal values of different WBCs

WBC	Percentage	Absolute value per cu mm
Neutrophils	50 to 70	3,000 to 6,000
Eosinophils	2 to 4	150 to 450
Basophils	0 to 1	0 to 100
Monocytes	2 to 6	200 to 600
Lymphocytes	20 to 30	1,500 to 2,700

LEUKOPOIESIS

Leukopoiesis is the development and maturation of leukocytes

**FIGURE 16.2: Leukopoiesis**

FUNCTIONS OF DIFFERENT WBCS ARE AS FOLLOWS

TABLE 16.5: Substances secreted by WBCs

WBC	Substance secreted	Action
Neutrophil	Proteases	Destruction of microorganisms
	Myeloperoxidases	
	Elastases	
	Metalloproteinases	
	Defensins	Antimicrobial action Anti-inflammatory action Wound healing Chemotaxis
	Cathelicidins	Antimicrobial action
	NADPH oxidase	Bactericidal action
	Platelet-activating factor	Aggregation of platelets
Eosinophil	Eosinophil peroxidase	Destruction of worms, bacteria and tumor cells
	Major basic protein	Destruction of worms
	Eosinophil cationic protein	Destruction of worms Neurotoxic action
	Eosinophil-derived neurotoxin	Neurotoxic action
	Interleukin-4 and 5	Acceleration of inflammatory response Destruction of invading organisms
Basophil	Heparin	Prevention of intravascular blood clotting
	Histamine	Production of acute hypersensitivity reactions
	Bradykinin	
	Serotonin	
	Proteases	Destruction of microorganisms
	Myeloperoxidases	
	Interleukin-4	Acceleration of inflammatory response Destruction of invading organisms
Monocyte	Interleukin-1	Acceleration of inflammatory response Destruction of invading organisms
	Colony stimulation factor	Formation of colony forming blastocytes
	Platelet-activating factor	Aggregation of platelets
	Chemokines	Chemotaxis
T lymphocytes	Interleukin-2, 4 and 5	Acceleration of inflammatory response Destruction of invading organisms Activation of T cells
	Gamma interferon	Stimulation of phagocytic actions of cytotoxic cells, macrophages and natural killer cells
	Lysosomal enzymes	Destruction of invading organisms
	Tumor necrosis factor	Necrosis of tumor Activation of immune system Promotion of inflammation
B lymphocytes	Immunoglobulins	Destruction of invading organisms
	Tumor necrosis factor	Necrosis of tumor Activation of immune system Acceleration of inflammatory response
	Chemokines	Chemotaxis

IMMUNITY

Immunity is defined as the capacity of blood to resist pathogenic agents. It is the ability of body to resist the entry of different types of foreign bodies like bacteria, virus, toxic substance etc.

Immunity is of two types :

1. Innate immunity
2. Acquired immunity

INNATE IMMUNITY

Also known as non-specific immunity. Present from the birth thus it is the inborn capacity of the body to resist pathogens. If any organism enters in the body, innate immunity eliminates them before development of any disease. This type of immunity represents first line of defence against any type of pathogens.

ACQUIRED IMMUNITY

Acquired immunity is the resistance developed in the body against any specific foreign body like bacteria, viruses, toxins, vaccines or transplanted tissues. So, this type of immunity is also known as specific immunity. It is the most powerful immune mechanism that protects the body from the invading organisms or toxic substances. Lymphocytes are responsible for acquired immunity.

Types of Acquired Immunity

Two types of acquired immunity develop in the body:

1. Cellular immunity – T lymphocytes
2. Humoral immunity- B lymphocytes

T-LYMPHOCYTES

T lymphocytes are processed in the thymus. Thymus secretes a hormone called thymosin which accelerates the proliferation and activation of lymphocytes in thymus and also it increases the activity of the lymphocytes in lymphoid tissue.

TYPES OF LYMPHOCYTES

During the processing, T lymphocytes are transformed into four types :

1. Helper T cells or inducer T cells. These cells are also called CD4 cells because of the presence of a molecule called CD4 on their surface.
2. Cytotoxic T cells or killer T cells. These cells are also called CD8 cells because of the presence of the molecule CD8 on their surface
3. Suppressor T cells
4. Memory T cells

STORAGE OF T LYMPHOCYTES

After the transformation, all the types of cells leave the thymus and are stored in lymphoid tissues of lymph nodes, spleen, bone marrow and GI tract.

B LYMPHOCYTES

B lymphocytes are first discovered in the bursa of the Fabricius in birds, hence the name B lymphocytes. Bursa of Fabricius is a lymphoid organ situated near the cloaca of the birds. Bursa is absent in mammals and the processing of B lymphocytes takes place in liver (during fetal life) and bone marrow (after birth).

TYPES OF B LYMPHOCYTES

After processing, the B lymphocytes are transformed into two types :

1. Plasma cells
2. Memory cells

STORAGE OF LYMPHOCYTES

After the transformation, B lymphocytes are stored in the lymphoid tissues of lymph nodes, spleen, bone marrow and the GI tract.

ANTIGENS

Antigens are the substances which induce specific immune reactions in the body. They are mostly conjugated proteins like lipoproteins, glycoproteins and nucleoproteins. Antigens are of two types :

1. Autoantigen or self-antigens present on the body's own cells such as A antigen, B antigen in RBCs
2. Foreign antigens or non-self-antigens than enter the body from outside.

NON-SELF/ FOREIGN ANTIGENS

Following are the non-self/ foreign antigens\:

1. Receptors on the cell membrane of microbial organism such as bacteria, viruses and fungi.
2. Toxins from microbial organisms
3. Material from transplanted organs or incompatible blood cells
4. Allergens or allergic substances like pollen grains

TYPES OF NON-SELF ANTIGENS

Depending upon the response developed against them in the body, the antigens are divided into two types :

1. Antigens, which induce the development of immunity or production of antibodies (immunogenicity)
2. Antigens, which react with specific antibodies and produce allergic reactions (allergic activity)

DEVELOPMENT OF CELL MEDIATED IMMUNITY

Cell mediated immunity is defined as the immunity developed by cell mediated response. It is also called cellular immunity or T cell immunity. It involves different types of cells such as T lymphocytes, macrophages and natural killer cells. Cell mediated immunity does not involve antibodies.

Cell mediated immunity is the major defence mechanism against infections from viruses, fungi and some bacteria such as tubercle bacillus. It is also responsible for delayed allergic reactions the rejection of the transplanted tissues.

Cell mediated immunity is offered by T lymphocytes and it starts developing when T lymphocytes come in contact with antigens. Usually, the microbial or non-microbial organism carry antigenic materials. These antigenic materials are presented to the helper T cells by antigen-presenting cells. Antigen-presenting cells are :

1. Macrophages - are the large phagocytic cells which digest the invading organism to release the antigen.
2. Dendritic cells - these cells are non-phagocytic in nature; they trap the antigens.

3. B lymphocytes - B lymphocytes also acts as antigen-presenting cells along with antigen receiving cells. However, they are least efficient antigen-presenting cells and need to be activated by helper T cells.

Antigen presenting cells present their antigens to the helper T cells which activates helper T cells through a sequence of events which are as follows:

1. Helper T cells recognizes the antigen displayed on the surface of the antigen-presenting cells with the help of its own surface receptor protein called T cell receptor.
2. Recognition of the antigen by the helper T cells initiates a complex interaction between the helper T cell receptor and the antigen. This reaction activated helper T cells
3. At the same time, macrophages release interleukin-1 which facilitates the activation and proliferation of helper T cells

Activated helper T cells enters into the circulation and activates all the other T cells and B cells.

CYTOTOXIC T CELLS which are activated by the helper T cells circulate through blood, lymph and lymphatic tissues and the destroy the invading organisms by attacking them directly. After finding the organism, the cytotoxic T cells enlarge and release cytotoxic substances such as lysosomal enzymes. These substances destroy the invading organisms. Cytotoxic T cells also destroys cancer cells, transplanted cells and any other cells which are foreign bodies.

SUPPRESSOR T CELLS also called regulatory T cells. These cells suppress the activities of the killer T cells. Thus, the suppressor T cells plays an important role in preventing the killer T cells from destroying the body's own tissues along with invaded organisms. Suppressor T cells also suppress the activities of helper T cells also.

Some of the T cells activated by an antigen do not enter the circulation but remain in lymphoid tissue, these T cells are called memory T cells. When the body is exposed to the same organism for the second time, the memory T cells identify the organism and immediately activates the other T cells. So, the invading organism is destroyed very quickly. The response of the T cells is also more powerful this time.

DEVELOPMENT OF HUMORAL IMMUNITY

Humoral immunity is defined as the immunity mediated by antibodies, which are secreted by B lymphocytes. B lymphocytes secrete the antibodies into the blood and lymph. Blood and lymph are the body fluids (humours or humors in Latin). Since B lymphocytes provide immunity through humor, this type of immunity is also known as humoral immunity.

Antibodies are the gamma globulins produced by B lymphocytes. These antibodies fight against the invading organisms. The humoral immunity is the major defence mechanism against the bacterial infection. Antigens presenting cells also activated B lymphocytes as the same way as it activates helper T cells. After activating B cells (B lymphocytes), proliferation begins and it transforms into two types of cells :

1. Plasma cells - plasma cells destroys the foreign organism by producing the antibodies. Antibodies are globulin in nature. Each plasma cells produce

about 2000 antibodies molecule per second. These antibodies are released into lymph and then transported into the circulation.

2. Memory cells - memory B cells occupy the lymphoid tissue throughout the body. The memory cells are in inactive condition until the body is exposed to the same organism for the second time. During the second exposure to the antigen, these memory cells produce antibodies at a faster rate. This phenomenon forms the basic principle of vaccination against the infections.

ANTIBODIES

An antibody is defined as a protein that is produced by the B lymphocytes in response to the presence of an antigens. Antibodies are gamma globulins in nature and it is also called immunoglobulins. Immunoglobins form 20% of the total plasma proteins. Antibodies almost enters all the tissues of the body.

TYPES OF ANTIBODIES

Five types of antibodies are identified :

1. IgA (Ig alpha)
2. IgD (Ig delta)
3. IgE (Ig epsilon)
4. IgG (Ig gamma)
5. IgM (Ig mu)

STRUCTURE OF ANTIBODIES

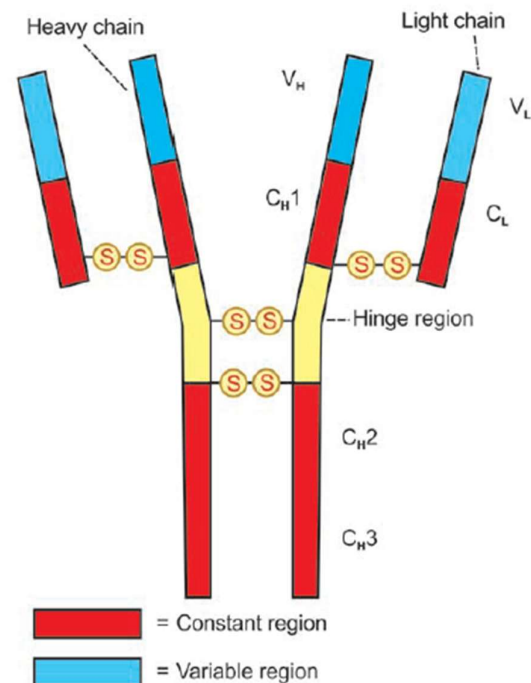
Antibodies are gamma globulins with molecular weight of 1,50,000 to 9,00,000. The antibodies are formed by two pairs of chains, namely one pair of heavy and one pair of light or short chains. Each heavy chains consist of 400 amino acids and each light chain consists of 200 amino acids.

Each antibody has two halves, which are identical. The two halves are held together by disulfide bonds (S-S). each half of the antibody is consists of one heavy chain (H) and one light chain (L). the two chains in each half is also joined by disulfide bonds (S-S). the disulfide bonds allow the movement of amino acid chains. In each antibody the light chain is parallel to one end of heavy chains. The light chain and the part of the heavy chain parallel to it form one arm. The remaining part of the heavy chains forms another arm. A hinge joins the both arms. Each chains of antibody includes two regions :

1. Constant region
2. Variable region

DIRECT ACTION OF ANTIBODIES

1. Agglutination - in this, the foreign bodies like RBCs or bacteria with antigens on their surfaces are held together in a clump by antibodies
2. Precipitation - in this, the soluble antigens like tetanus toxin are converted into insoluble form and then precipitated.



3. Neutralization - during this, the antibodies covers the toxic sites of antigenic products
4. Lysis - it is done by the most potent antibodies. These antibodies rupture the cell membrane of the organism and then destroy them.

IMMUNIZATION

Immunization is defined as the procedure by which the body is prepared to fight against a specific disease. It is used to induce the immune resistance of the body to a specific disease. Immunization are two types :

1. Passive immunization - passive immunization or immunity is produced without challenging the immune system of the body. It is done by administration of serum or gamma globulin from a person who is already immunized to a non-immune person. Passive immunization is acquired either naturally or artificially.
2. Active immunization - active immunization or immunity acquired by activating immune system of the body. Body develops resistance against disease by producing antibodies following the exposure to antigens. Active immunity is acquired either naturally or artificially.

AUTOIMMUNE DISEASE

Autoimmune disease is defined as a condition in which the immune system mistakenly attacks the body's own cells and tissues. Normally an antigen induces the immune response in the body. The condition in which immune system fails to give response to an antigen is called tolerance. Normally the body has tolerance against self antigen. However, in some conditions , the tolerance fails or becomes incomplete against self-antigen. This stage is called autoimmunity and it leads to the activation of T lymphocytes or production of antibodies from B lymphocytes. The T lymphocytes or autoantibodies attack the body's normal cells whose surface contains self-antigen or autoantigen. Common autoimmune diseases are :

1. Insulin independent diabetes mellitus
2. Myasthenia gravis
3. Hashimoto thyroiditis
4. Grave's disease
5. Rheumatoid arthritis

PLATELETS

Platelets or thrombocytes are the formed elements of the blood. Platelets are small, colorless, non-nucleated and moderate refractive bodies.

Diameter : 2.5 μ

Volume : 7.5 cu μ

FUNCTION OF PLATELETS

1. Role in blood clotting - platelets are responsible for the formation of intrinsic prothrombin activator.
2. Role in clot retraction - in blood clotting, blood cells including platelets are entrapped in between the fibrin threads, cytoplasm of the platelets contains the contractile proteins, namely actin, myosin and thrombosthenin, which are responsible for the clot retraction.

3. Role in prevention of the blood loss - due to adhesive property of the platelets, they seal the damage in blood vessels like capillaries.
4. Role in repair of ruptured blood vessels - platelet-derived growth factor (PDGF) formed in cytoplasm of platelets is useful for the repair of the endothelium and other structures of the ruptured blood vessels
5. Role in defence mechanism - by the property of agglutination, platelets encircle the foreign bodies and destroy them.

PLATELET DISORDERS

1. Thrombocytopenia - decrease in platelet count
2. Thrombocytosis - increase in platelet count
3. Thrombocythemia - is a condition with persistent and abnormal increase in platelet count
4. Glanzmann's thrombasthenia - it is an inherited hemorrhagic disorder, caused by structural or functional abnormality of platelets. It leads to thrombasthenic purpura. However, the platelet count is normal. It is characterized by normal clotting time, normal or prolonged bleeding time but defective clot retraction.

COAGULATION OF BLOOD

Coagulation or clotting is defined as the process in which blood loses its fluidity and becomes jelly like mass few minutes after it is shed out or collected in a container.

FACTORS INVOLVED IN BLOOD CLOTTING

13 factors are identified :

- | | |
|-------------|--|
| Factor I | : fibrinogen |
| Factor II | : prothrombin |
| Factor III | : thromboplastin (tissue factor) |
| Factor IV | : calcium |
| Factor V | : labile factor (proaccelerin or accelerator globulin) |
| Factor VI | : presence has not been proved |
| Factor VII | : stable factor |
| Factor VIII | : antihemophilic factor (antihemophilic globulin) |
| Factor IX | : Christmas factor |
| Factor X | : stuart-prower factor |
| Factor XI | : plasma thromboplastin antecedent |
| Factor XII | : Hageman factor (contact factor) |
| Factor XIII | : fibrin-stabilizing factor (fibrinase) |

STAGES OF BLOOD CLOTTING

In general blood clotting occurs in three stages:

1. Formation of prothrombin activator - blood clotting commences with the formation of substance called prothrombin activator which converts prothrombin into thrombin its formation is initiated by the substance produced either within the blood or outside the blood. Thus, formation of prothrombin activator occurs in two pathways:
 - a. Intrinsic pathway - the formation of prothrombin activator is initiated by platelets, which are within the blood itself.
 - b. Extrinsic pathway - formation of prothrombin activator is initiated by the tissue thromboplastin, which is formed from injured tissues.

2. Conversion of prothrombin into thrombin
3. Conversion of fibrinogen into fibrin

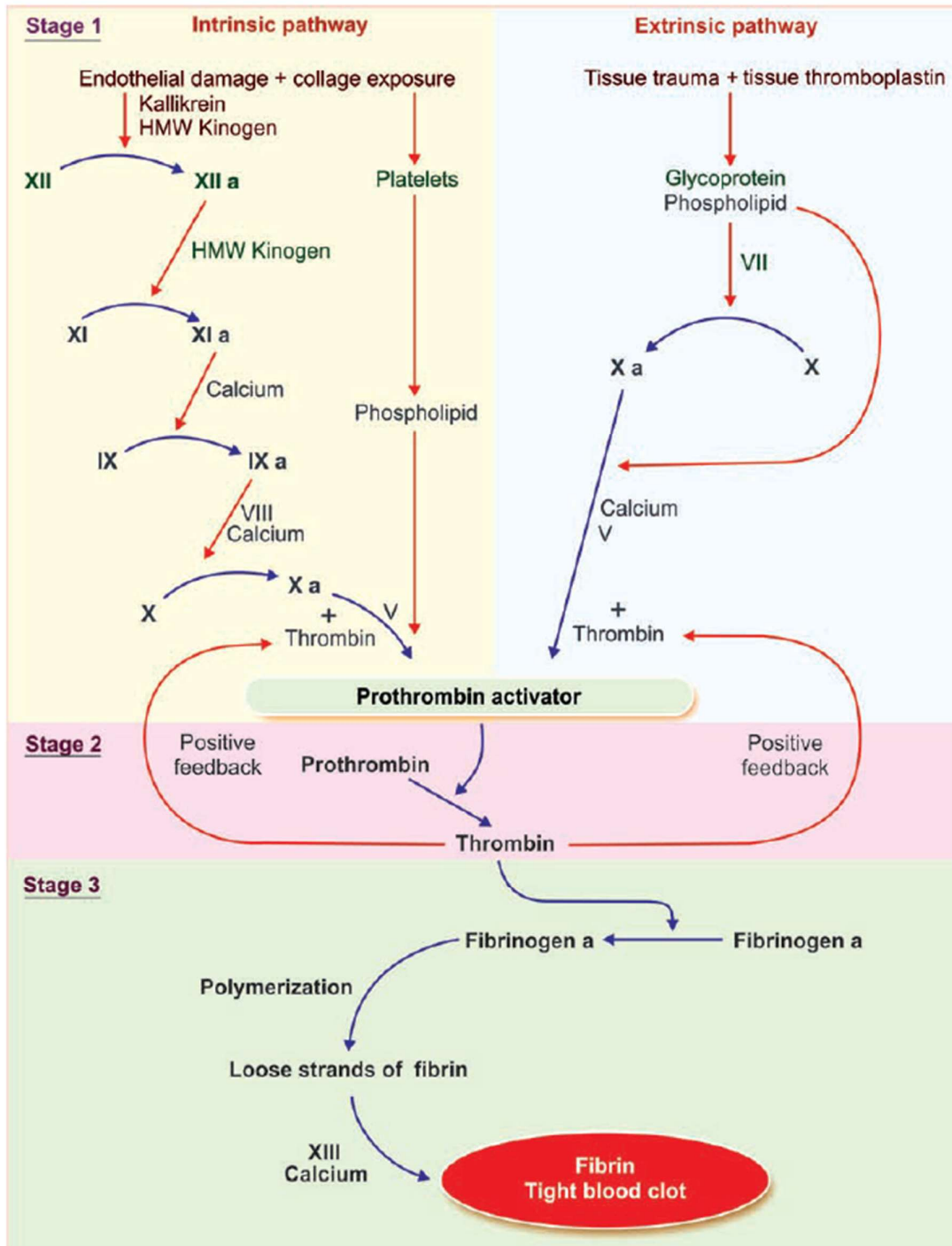


FIGURE 20.1: Stages of blood coagulation. a = Activated, + = Thrombin induces formation of more thrombin (positive feedback); HMW = High molecular weight.

BLOOD CLOT

Blood clot is defined as the mass of coagulated blood which contains RBCs, WBCs and platelets entrapped in fibrin meshwork. The RBC and WBC are not necessary for clotting process, however when clot is formed, these cells are trapped in it along with platelets. The trapped RBCs are responsible for the red color of the clot. The external blood clot is called scab. It adheres to the opening of damaged vessel and prevent blood loss.

CLOT RETRACTION

After the formation, the blood clot starts contracting and after about 30 to 45 minutes, the straw-colored serum oozes out of the clot and the oozing of the serum is called clot retraction. Contractile proteins, namely actin, myosin and thrombosthenin in the cytoplasm of platelets are responsible for clot retraction.

ANTICOAGULANTS

Anticoagulants are the substances which prevent or postpone coagulation of blood
HEPARIN

Heparin is a naturally produced anticoagulant in the body. It is produced by mast cells which are the wandering cells present immediately outside the capillaries in many tissues and organs that contain more connective tissue. These cells are abundant in liver and lungs. Basophils also secrete heparin

Heparin prevents the blood clotting by its antithrombin activity. It directly suppresses the activity of thrombin

BLOOD GROUPS

More than 20 genetically determined blood group system is known today. But Karl Landsteiner discovered two blood group systems called ABO system and RH system.

ABO SYSTEM

Based on the presence and absence of antigen A and antigen B, blood is divided into four groups:

1. 'A' group
2. 'B' group
3. 'AB' group
4. 'O' group

TABLE 21.1: Antigen and antibody present in ABO blood groups

Group	Antigen in RBC	Antibody in serum
A	A	Anti-B (β)
B	B	Anti-A (α)
AB	A and B	No antibody
O	No antigen	Anti-A and Anti-B

LANDSTEINER LAW

Landsteiner law states that:

1. If a particular agglutinin (antigen) is present in the RBCs, corresponding agglutinin (antibody) must be absent in the serum
2. If a particular agglutinin is absent in the RBCs the corresponding agglutinins must be present in the serum.